CHAPTER V
Epistemological implications in Artificial Intelligence

Indian

In the era of computers of the fifth generation, there has been a significant amount of work devoted into the simulation of intelligent behaviour in machines. By using grammar-based computing, in which the Knowledge of Language (KOL) is sought to be given to machines for Natural Language Understanding (NLU) or Processing (NLP), and by using knowledge-based computing, in which the background/contextual or Real World Knowledge (RWK) is sought to be represented in machine for understanding real world situations and problem solving, the researchers are attempting to find a solution to this problem. Both of these approaches are being considered by the researchers. An increasing number of people are coming to the realisation that RWK representation is necessary for the simulation of consciousness in machines. With that being said, there has not been a single framework that has been able to be developed that is applicable to all sorts of knowledge, identities of the locus, object, and instrument of cognition. Many academics in the field of artificial intelligence are beginning to question whether or not the following statements are true:

* Knowledge can be represented
* Computers can encode and decode knowledge in everyday language
* Computers are capable of thinking and understanding
* Computers require knowledge in order to communicate with people or other machines

Although Panini's grammar is certainly applicable to RWK, it is actually an account of KOL that can be used as a model for other languages as well. This is because there are indisputable descriptions of both the KOL and the RWK in India, particularly in the Sanskrit language. It is possible to find one of the first accounts of RWK in the NN system, which is not too dissimilar to the accounts that the AL philosophers have provided. As a result of this, it is of the utmost importance to investigate what the NN system has to offer. It is to compare and contrast the Indian epistemological tradition, specifically the NN, with the one that is associated with artificial intelligence.

The western perspective on knowledge has traditionally centred on the question of whether knowledge is learned (posteriori) or given (a priori), which means "from what is before" instead of "from what is after." In contrast to the latter, which developed into the empiricist way of inquiry, the former ultimately evolved into the rationalistic mode of inquiry. Plato contends that there is a consciousness of absolute, universal concepts or forms that exist apart from any

individual effort to comprehend them. He says this awareness exists independently of any subject. In his approach to the acquisition of knowledge, Aristotle places a greater focus on logical and empirical approaches. Moreover, he acknowledges the viewpoint that knowledge is an understanding of fundamental truths that are universally applicable. All of the early rationalists, including Descartes, Spinoza, and Leibniz, believed that the most important source of knowledge was deductive reasoning that was founded on axioms. The early empiricists, such as Francis Bacon and John Locke, considered sense perception to be the primary source of information and the ultimate test of knowledge. Despite the fact that Berkeley shared Locke's views on the relationship between knowledge and ideas, he was opposed to the idea that ideas should be separated from objects. As an empiricist, Hume was not in agreement with Berkeley's conclusion that knowledge may be derived solely from philosophical concepts. Hume classified all knowledge into two categories: relations of concepts and the knowledge that can be found in mathematics and logic. This knowledge is precise and certain, but it does not convey any information of any type about the world. the knowledge that is obtained from the senses and the facts that are known. The debate between empiricists, rationalists, pragmatists, and constructivists persisted both throughout the Renaissance and after it had ended. The reflection-correspondence hypothesis, which is part of the empiricist school of thought, asserts that knowledge is the product of a reflection of external things onto our brain or mind throughout the learning process. It is their belief that the meaning is what the words signify, and that language, much like a mirror, reflects the things that are going on around us. If there is a correspondence between the relations between the representations of things and the relations between the things themselves, then the assertion stated is true. An indication of untruth is a reflection that is incorrect and distorted, which is a picture that does not correlate to any real thing in the world. Kant was the one who sought to develop a synthesis of the competing arguments that were presented by rationalists and empiricists. Knowledge is said to be the consequence of the organising of perceptual facts on the basis of inborn cognitive structures or categories, as stated by Kant. Space, time, things, and causality are all items that fall under the Kantian category. According to Distantie (2003), Kant differentiated between three distinct categories of knowledge.

Kant, Spencer, Bergson, Alexander, and Dawes Hicks are some of the philosophers who are credited with being the pioneers of the Act Theory of Knowledge. This theory attempts to explain knowledge in terms of activity. The behaviourists, on the other hand, refer to it as an activity of the body; nevertheless, contemporary European philosophers such as Moore and Broad have not accepted this idea. The Self-Subsistency Theory, which asserts that knowledge

is something that exists on its own, is supposed to have been proposed by Berkeley, who is considered to be the most prominent western philosopher (Bijalwan 1987).

There are two major fields of research in artificial intelligence: knowledge representation (KR) and information retrieval (IR). Knowledge can be characterised as objects and attributes, or even as an algorithm that contains objects and procedures, when approached from the perspective of artificial intelligence (AI). An interaction between "data" and "processes" or "abilities" and "memory storage" is stated to be the cause of the so-called "intelligent behaviour" that is observed. According to Jean-Raymond Abrial (1974), a database is a representation of the physical world as it is always changing. During a particular moment, the state of this model is a representation of the knowledge that it has obtained from the outside world. But knowledge is more than static encoding of facts in interacting with the world. The idea that knowledge of anything is the capacity to construct a mental model that accurately depicts the thing in question, as well as the actions that may be carried out by it and on it, is one of the fundamental premises of artificial intelligence (AI). Following this, individuals (or robots) are able to make predictions about what is likely to occur in the real world by testing actions on the model (Sowa in 1984). An earlier challenge faced by those working on artificial intelligence was determining whether to represent knowledge by using a declarative approach or a procedural method. For the purpose of knowledge representation, it has been finally decided that the most appropriate strategy would be one that combines a declarative approach with a procedural approach. Predicate calculus, semantic nets, frames, conceptual graphs, scripts, semantic primitives, and neural nets are some of the other well-known techniques for KR. These techniques follow a wide variety of strategies for KR. In spite of this, reducing the unknown phenomenon known as knowledge to a single or two definitions would not be an aim in and of itself. According to Hopcroft and Lipman (1979), not only would it be necessary to comprehend the entirety of the human cognitive process, but it would also be necessary to develop efficient methods for representing information.

pratyaksa (perception)

* anumana (inference)

* upamana (analogy)

* sabda (verbal testimony) and latter is sub-classified as

* samsaya (doubt)

* viparyaya (error)

* tarka (hypothetical argument)

* smrti (memory)

5.1. **Comparison with AI**

The following sub-sections define each of the above sub-classifications of invalid and valid

knowledge and give their AI parallels--

5.1.1. apramana--the invalid knowledge

It is defined as false knowledge, doubtful or illusionary. The four types of invalid knowledge are

defined and compared with AI as follows:

5.1.1.1. smrti (memory)

* is impressions left by previous knowledge

* it is non valid because the object of knowledge is absent at the time of remembrance

* AI explanation

* memory as gaps in knowledge or probabilistic knowledge which must be corrected before

processing starts and conclusion is arrived at

* such gaps or incomplete knowledge may be replaced by new knowledge [right arrow] nonmonotonic knowledge (McDermott & Doyle 1980)

* static memory [right arrow] knowledge of object (WHAT)

* short term : memory in current focus

* long term: memory for future use

* procedural memory [right arrow] knowledge of behavior (HOW)

* short term

* long term

5.1.1.2. samsaya (d * conflicting judgment on the precise character of an object of cognition (Bijalwan 1987)

* neither true or false, only wavering judgment

* AI explanation

* ambiguity resolution and bug handling come close to samsya

5.1.1.3. tarka (hypothetical judgment)

* AI explanation

* Heuristics

* when reality is not known, truth is ascertained trying alternatives

* atmasraya (petito principi)

* if A [right arrow] A then A must be different from A

* anyonyasraya (dependent tarka)

* A[right arrow]B[right arrow]A

* cakraka (circular logic)

* recursion in AI

* A[right arrow]B[right arrow]C[right arrow]A

* anavastha (regressus ad infinitum or transitivity)

* A[right arrow]B[right arrow]C[right arrow]D ...

* tadanya-budhiat-prasanga (reduction ad absurdum)

* proves a proposition by negating the contradiction of the proposition

* A[right arrow]-(-A) or + [right arrow] -(-)oubt)

5.1.1.4. viparyaya (error)

* manifestation of a real object in another * positive misconception, incomplete knowledge, gaps in knowledge * AI explanation * can be compared to bugs and inconsistencies

5.1.2. pramana--the valid knowledge

The following section discusses the means to valid knowledge and compares them with AI

5.1.2.1. **pratyaksa (perception)**

* Five senses--sight, smell, taste, hearing, touch (mind [right arrow] sixth sense for cognition of

feelings like pleasure, pain etc)

* six ways of perception for six kinds of objects

* substance, quality, universal quality, sound, sound-ness, nonexistence

* AI [right arrow] icon, schema and percept

* icon [right arrow] temporary record of a sensory input that the brain keeps

* schema [right arrow] blue print of the mental model

* percept [right arrow] assembling units which are to be assembled according to the schema to

create a mental model

* Sowa (1984) calls the search mechanism "associative comparator" which must have the following

characteristics:

* associative retrieval

* brain is like a computer, retrieves data by an address in the storage by matching the best pattern

* top-down match

* perception searches for percepts that match the overall pattern of an icon

* stimulus constancy

* stimuli from same external object are recognized as equivalent despite varying size, brightness etc

* distributed storage

* images are not stored at a specific point, but distributed

5**.1.2.2. anumana (inference)**

* Gautama presupposes perception

* Jayanta gives five processes of inferring

* Perception of reason

* Remembrance of the universal concomitance

* Judgment that the subject of inference contains sense which, is concomitant with the object

* Knowledge of the consequence, and

* Judgment that the consequence is worthy of being accepted or rejected

* AI

* inference is very important and strong point of predicate calculus

* axioms [right arrow] theorems

* new facts can be deduced from axioms using rules of inference similar to ones given by Gautama

Gautama gives 3 ways to infer:

* psrvavat (deduction)

* cause [right arrow] effect.

* cloud [right arrow] rain

* "Gauri eats grass" from "Gauri is a cow" and "all cows eat grass" --

(inst gauri cow)

(forall X (if (inst X cow) (food X grass)))

(food gauri grass)

* sesavat (abduction)

* process [right arrow] explanations

* effect [right arrow] cause

* river is swollen [right arrow] there could have been rain

* Abduction has the following paradigms

From: b

(if a b)

Infer: a

* unlike deduction, abduction is not a legal interference. It can lead to false conclusions--

from: (feels nervous Ira)

(forall (X) (if (is sick X)

(feels nervous X)))

infer: (is sick Ira)

* samanyatod (induction)

* infers consequent from the antecedent, which is neither cause nor effect

* while induction can take several forms, the most common is

from: (P a), (P b),...

infer: (forall (X) (P X))

* although this is not a sound inference, induction like abduction, is very useful in everyday life where

it is more commonly known as learning

* for example, if we see a lot of leaves of green color, we might infer that all leaves are of green color

from: (if(inst leaf-1 leaf) (colour leaf_1 green))

(if(inst leaf-2 leaf) (colour leaf_2 green))

Infer: (forall (X) (if(inst X leaf) (colour X green)))

* Vacaspati divides inference into vita and avita (Bijalwan 1987). The former is based on universal

agreement in presence. As an example, we can infer

from: whatever has smoke has fire the hill has smoke

infer: the hill has fire

* avita is based on universal agreement in absence

from: whatever is non-different from other objects has no

smell. Earth has smell

infer: Earth is different from others.

* Prasastapada has svarthanumana and pararthanumana as two types of inference.

* Svarthanumana

* the premises are known from our own experience

*pararthanumana

* They are discovered by one man and imparted into another through language.

* Therefore, there is greater chance of error in pararthanumana which is based on words.svarthanumana is a mental process which is divided into data and samanyatodata by Prasastapada(Bijalwan 1987).

* Udyotakara gives kevalanvayin (K), kevalavyatirekin (KV), and amayavyatirekin (AV) as three types of inference. If a middle term is only positively related to the major term it is called K. Here. Thereason exists in the subject and similar instances, and is devoid of dissimilar instances. For example--we can infer "pot is nameable" from "pot is knowable" and "all knowable things are nameable"--

(property X nameable)))

(if (inst pot knowable)

(property pot nameable))

(property pot nameable)

In the case of KV, the middle term is negatively related to the major term. Reason exists in the

subject, but not in dissimilar instances. For example--

from: (no (not P) is (M))

S is M

infer: S is P

When the middle is positively as well as negatively related to the major term, and reason exists in the entire subject, and in all similar instances but does not exist in dissimilar instances, the case is that of AV, as "sound is non-eternal because it is produced like a jar". The inference of Fire from smoke is also of this kind. Jayanta does not however, accept this division of inference in to K, KVand AV by Udyotakara (Bijalwan 1987).

5.1.2.3. upamana (analogy)

* upamana is the means by which we gain the knowledge of a previously unknown object on thebasis of its similarity to another object previously well known

* Mimansists and the Vedantists also accept analogy as an independent source of knowledge

* AI

* analogy is too slippery to be seen as just another road to induction

* often, analogy matches are so slipshod that they merely suggest solutions to the problem ofknowledge acquisition in general, and for inductive learning in particular

* analogy is useful because it serves as initial framework for developing the new concepts

5.1.2.4. sabda (verbal testimony)

* Leaving aside the Carvakas, Vaisesikas and the Buddhists, all the other systems of Indian philosophy accept sabda as a distinct source of knowledge, but they also differ with one another with respectto its nature forms and a number of other aspects.

* Jayanta (Nyayamanjari)

* contains a long discussion on this problem

* justices the acceptance of sabda as a distinct means of knowledge

* gives serious thought to the philosophy of language

* tries to explain the views of his predecessors and evaluates

* the arguments of prominent scholars

* his account of sabda is as much relevant to grammar, rhetoric, and linguistics as it is to logic

* convincingly proves that words do not exist before their production

* also refutes the sphota theory of grammarians and argues that the relationships between words and meaning is conventional

* AI

* sabda can be compared to the concept of formal string and language.

* language is a set of strings made from some alphabet

if a fixed alphabet = E, then the language formed by

E is called [E.sup.*]

ifE= {a} then

[E.sup.*] = {e. a. aa. aaa. ...}

Both the Naya system and the AI system are subject to comparison and contrast. As has been demonstrated, the Naya system shares many similarities with the AI in terms of how it handles valid and invalid kinds of knowledge. The AI, on the other hand, employs a variety of different methods in addition to logic and inference. These methods, including as semantic networks, conceptual graphs, frames, rules-based systems, and processes, offer suggestions for practical ways to represent information. This is in contrast to the NN and even western schools, which place an excessive amount of focus on fixing the identity of the locus of knowledge acquisition. The latter is perhaps not as important for artificial intelligence at the moment. In spite of this, there is a pressing need to conduct additional research into a number of Indian systems, such

as the neural network (NN) system and others, such as Buddhist logic, in order to develop appropriate models of human cognition that can be utilised by artificial intelligence.
The way in which Nyāya differentiates between valid and invalid information is similar to the way in which artificial intelligence differentiates between accurate forecasts and errors. False inference, also known as mithyājñāna, may be compared to algorithmic bias or misclassification in artificial intelligence (AI). In the Nyāya approach, errors are corrected by reasoning, which is similar to the training of artificial intelligence models, where recurrent learning helps reduce errors. For the purpose of verifying truth claims, artificial intelligence could utilise logical validation in the style of Nyāya.I.e. New artificial intelligence mistake-detection frameworks can be inspired by Nyaya's classification of perceptual and inferential errors, which is used in error analysis. There are similarities between the hierarchical knowledge graphs in artificial intelligence and the structured approach to cognition that Nyāya takes. Co-Reasoning between Humans and Artificial Intelligence: The approach of combining human logical insight (Nyāya) with machine inference (AI) has the potential to improve interpretability and accountability. The logical structure of Nyāya places a strong focus on the cause-effect link, also known as vyāpti, which states that a particular cause consistently results in a particular effect. For instance, smoke suggests fire. When it comes to artificial intelligence (AI), this is comparable to the concept of causal reasoning and explainable AI (XAI), which involves ensuring that the output of an AI is not only statistically accurate but also logically and causally warranted. This idea is at the core of Nyāya logic.

When comparing the Nyāya–Nyāya (NN) system with the artificial intelligence (AI) systems of today, it is important to note that each system approaches knowledge differently. It has been demonstrated that the neural network (NN) system has a number of significant similarities with artificial intelligence (AI), particularly in its systematic differentiation between legitimate (pramā) and invalid (aprāmā) kinds of knowledge. The two frameworks acknowledge that knowledge is not simply the accumulation of information; rather, it must fulfil certain characteristics of trustworthiness, correctness, and justification in order to be considered knowledge as such. On the other hand, whereas the NN system is primarily dependent on organised logical inference and properly stated epistemic categories, contemporary AI systems make use of a wider variety of approaches that go beyond the traditional methods of logic and inference alone.

With that being said, the significance of Indian epistemological frameworks for artificial

intelligence research is not diminished in any way. To the contrary, there is still a significant requirement to engage with Indian philosophical traditions, such as the NN system and Buddhist logic, in a more in-depth manner in order to develop models of cognition that are more complex and subtle. For example, Buddhist theories of inference, momentariness, and dependent origination provide alternate methods of explaining reasoning processes. These theories do not presuppose the existence of a permanent self, which is consistent with the non-subjective character of artificial intelligence systems. Likewise, the rigorous examination of mistake, doubt, and verification that the NN system does can provide valuable insights for the development of artificial intelligence systems that are more resilient, transparent, and epistemically accountable.

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The status of machine knowledge is a significant epistemological conclusion that arises from artificial intelligence. Understanding knowledge as a human effort that involves belief, truth, and justification has been the conventional understanding of the concept. It is possible for artificial intelligence systems to store knowledge, draw conclusions, learn from data, and provide predictions. In light of this, the question arises as to whether artificial intelligence systems actually know things or only manipulate symbols and figures. People who adhere to a strict epistemological perspective contend that artificial intelligence (AI) does not contain understanding and consciousness, and as a result, it does not possess knowledge in the full human sense, but rather only derived or instrumental knowledge.

Moreover, artificial intelligence reshapes disputes concerning the origins of knowledge. Knowledge that is a priori, which is independent of experience, and knowledge that is a posteriori, which is gained from experience, are two types of knowledge that are distinguished by classical epistemology. Models of knowledge acquisition that are empiricist in nature are similar to machine learning systems since they rely largely on data and experience. A priori, on the other hand, refers to the nature of the algorithms, logical frameworks, and mathematical models that guide artificial intelligence. Artificial intelligence (AI) blurs the traditional epistemic boundaries by combining aspects of rationalist and empiricist approaches.

Another crucial implication is to the trustworthiness and explanation of the statement. Reasons, evidence, and critical reflection are the means by which human epistemology seeks to justify views. Particularly when it comes to deep learning models, artificial intelligence systems frequently generate outputs that are extremely accurate yet difficult to explain. Epistemic opacity is a problem that arises as a result of this, in which even developers are unable to completely account for how an artificial intelligence arrived at a conclusion. As a consequence of this, concerns arise regarding the epistemological justification of beliefs or recommendations that are formed by means of artificial intelligence.

Additionally, artificial intelligence brings about new kinds of epistemic authority and trust. AI systems are having an increasingly significant impact on decision-making in a variety of sectors, including medical, law, economics, and research. This causes the epistemic responsibility to move from the individual human knowers to the complex socio-technical systems around them. It is now necessary for epistemologists to enquire about who is accountable for the knowledge claims that are generated by artificial intelligence: the system, the programmer, the data, or the user, and how trust ought to be allocated.

The development of artificial intelligence has an additional impact on the boundaries of knowledge. Artificial intelligence has the ability to comprehend enormous datasets that are beyond the cognitive capabilities of people, which enables new discoveries and patterns that humans could never see. Artificial intelligence is, however, constrained by factors such as the quality of the data, biases, and the architecture of the algorithm. These limits serve as a reminder that the information behind artificial intelligence is neither objective or flawless; rather, it is formed by human values and assumptions that are entrenched in its development.

When it comes to social and collective epistemology, artificial intelligence has ramifications. Collaborations between humans and machines are making an increasing contribution to the production of knowledge. The significance of ideas like distributed cognition, extended mind, and common knowledge is beginning to regain its previous prominence. The incorporation of artificial intelligence into the epistemic environment brings about a transformation in the manner in which societies generate, validate, and distribute knowledge.

The fact that identity theory is dubious of powerful artificial intelligence is a significant implication that it has for the field of artificial intelligence. If some mental states are the same as certain biological brain states, then it is impossible for computers made of silicon, circuitry, and code to actually possess mental states. This is because computers are not capable of developing their own mental states. The simulation of intelligence or awareness by artificial

intelligence systems is a possibility; nevertheless, from this perspective, it is important to note that these systems do not genuinely think or feel. The conclusion that might be drawn from this is that the notion that artificial intelligence can ever have authentic feelings, experiences, or beliefs is called into question.

Furthermore, identity theory places a considerable amount of emphasis on the biological underpinnings of cognitive processes. This argues that in order to acquire true intelligence through the field of artificial intelligence research, it would be required to design systems that could closely duplicate the physical structure and activities of the human brain. This would be necessary in order to reach genuine intelligence. Models of computing that are simple or algorithms that are abstract would not be sufficient in this situation. Furthermore, as a result of this, identity theory offers philosophical support to a number of approaches, such as artificial intelligence that is inspired by neurological processes, brain emulation, and biologically realistic neural modelling.

the limitation that it imposes on the concept of multiple realisability through its implementation. In contrast to functionalism, which allows mental states to be realised in a broad range of other physical systems, identity theory restricts mental states to particular subtypes of physical states, notably those that are associated with the human brain. The upshot of this is that identity theorists may argue that artificial intelligence does not possess true consciousness since it does not have the same physical identity as a human brain, even if it behaves exactly like a person at all times. This is because artificial intelligence does not share the same physical properties as a human brain.

On the other hand, identity theory does not in any way undermine the practical applications of artificial intelligence in the real field. From this perspective, artificial intelligence (AI) can still be of immense use as a tool for replicating human cognition, anticipating behaviour, and offering support with decision-making. These are all areas in which AI can be extremely beneficial. Artificial intelligence models have the potential to supply neuroscientists with simulations that can be tested, which can be of great assistance to them in comprehending how the operations of the brain function. Despite the fact that the models themselves are not conscious, this is nevertheless the case.

Because it argued that intelligence can be linked with behaviour rather than with the mental processes that occur within an individual, behaviourism had a significant impact on the early concept of artificial intelligence (AI). This was due to the fact that behaviourism proposed that intelligence can be linked with behaviour. The behaviourist school of thinking holds that an

artificial system can be considered intelligent if it exhibits behaviours that are indistinguishable from those of a human being. This is because the behaviour of an artificial system is indistinguishable from that of a human being. Alan Turing's Turing Test is a test that examines the intelligence of computers solely on the basis of observable language behaviour, without making any reference to the mental states of the machines themselves. There is a significant connection between this notion and the Turing Test, which is a test that analyses the intelligence of machines. Therefore, behaviourism provided a conceptual justification for evaluating artificial intelligence only on the basis of its performance, rather than on the basis of its fundamental architecture.

In the field of artificial intelligence, behaviourism is characterised by the fact that it focuses a strong emphasis on the connections between input and output. In the early stages of the development of artificial intelligence systems and machine learning models, the stimulus–response framework that is a part of behaviourist psychology was taken into consideration. It was the intention of these systems to provide appropriate outputs in response to certain inputs. In particular, reinforcement learning is a form of learning that is reflective of behaviourist notions. This type of learning is characterised by the use of both rewards and penalties in order to shape the behaviour of an agent. In this sense, behaviourism played a significant role in setting the groundwork for the development of realistic artificial intelligence systems that are based on learning from its surroundings through feedback.

Behaviourism also conveys the suggestion that the existence of internal mental states is not necessary for intelligence to exist. For an artificial intelligence system to be considered intelligent, it is not required for it to have consciousness, comprehension, or subjective experience; rather, it is sufficient for it to behave in a manner that is consistent with intelligent behaviour. Weak artificial intelligence is a type of artificial intelligence in which machines simulate intelligent conduct without claiming true comprehension or intellect. This provides support to the establishment of weak AI, which is a sort of AI. As a consequence of this, behaviourism avoids engaging in complex metaphysical debates concerning the presence of mind and consciousness in machines.

The behaviourist approach to artificial intelligence, on the other hand, has consistently been the target of severe criticism. The argument that engaging in intelligent activity does not necessarily indicate true comprehension is one of the most major concerns that comes up. John Searle's argument in the Chinese Room poses a challenge to the behaviourist theory since it demonstrates that a system can create correct responses without fully knowing the meaning of those responses. Some individuals, on the other hand, are of the opinion that behaviourism does

not take into consideration internal cognitive processes like as representation, reasoning, and intentionality. These processes are essential for offering an explanation for higher-level intelligence.

Regarding the ethics of artificial intelligence, there is also a substantial impact. Artificial intelligence systems are unable to have internal experiences such as joy or anguish, and as a result, they are unable to sense moral damage. This is based on the assumption that behaviourism is accurate. As a result of this, artificial intelligence systems do not have any moral position, and they are completely responsible for ethical behaviour. The responsibility for ethical behaviour resides with humans. Despite the fact that this makes it simpler to make ethical decisions, it may become troublesome in the future when artificial intelligence systems behave in a manner that is more like to that of human beings.

The conclusion is that behaviourism has been a significant influence on the creation of artificial intelligence as well as the appraisal of its effectiveness. Because behaviourism lays an emphasis on visible behaviour, performance, and learning through reinforcement, this is the reason why this is the case. It was a valuable and significant paradigm for early research on artificial intelligence; nevertheless, the fact that it did not take into consideration the mental processes that occur within the mind inhibits its capacity to describe genuine knowledge and consciousness. This is despite the fact that it was a helpful and important paradigm. Although more complex cognitive and functional theories have taken their place, behaviourism continues to be an important historical and philosophical grounding for artificial intelligence (AI) as it continues to grow. This is the case despite the fact that these theories have taken their place.

In the field of Artificial Intelligence (AI), functionalism has been an important and influential feature in both the research and application of AI. Functionalism provides a significant conceptual foundation for artificial intelligence (AI) by advocating for the idea of multiple realisability. It is feasible to achieve the same mental state in a variety of different physical systems by utilising this method. Some examples of these physical systems include the human brain, the neurological systems of animals, and even artificial robots. As a result of this, researchers working in the field of artificial intelligence do not have to imitate the biological structure of the human brain in order to produce intelligent systems through their work. These individuals, on the other hand, focus their attention on replicating the functional organization of cognition, which is in complete accordance with the functionalist philosophical perspective. One of the oldest applications of functionalism in the field of artificial intelligence is symbolic or classical artificial intelligence, which is an example. The early models of artificial

intelligence were based on machine-state functionalism, which served as their primary origin of inspiration. According to these conceptions, the mind is a computer system, and the mental operations that occur are considered to be computational states. Artificial intelligence programs made use of symbols, rules, and algorithms in order to communicate their beliefs, goals, and intents to the public. In addition to these symbolic representations, functional activities such as thinking, planning, and problem-solving were developed during the process of development. This was done in order to reflect the functionalist view that mental states are determined by the causal functions that they perform within a system. This was done in order to indicate their significance.

Additionally, the rise of cognitive architectures such as SOAR (State, Operator, and Result) and ACT-R and (Adaptive Control of Thought—Rational) is evidence that functionalism is actively present. These systems seek to imitate human cognition by disassembling the mind into its component elements, which are functionally distinct from one another. These components include perception, memory, learning, and action, among others. What characterises a component is not the manner in which it is physically manifested; rather, it is the function that it performs that is the defining characteristic of the component. In a similar manner, this reflects the functionalist position, which maintains that mental processes may be characterised by the roles that they play within an overall cognitive system.

Artificial intelligence in the modern era is progressively incorporating functionalism through the utilisation of artificial neural networks and deep learning systems. In spite of the fact that they are subject to a significant amount of impact from biological neurones, these networks do not correspond to the physical structure of the brain. They instead carry out cognitive activities by means of layers of interconnected units that are responsible for simultaneously processing inputs, transforming internal states, and producing outputs. These units are responsible for carrying out these tasks simultaneously. It is not the biological parallels between these systems that are relevant; rather, it is the functional performance of these systems that is important; this is what makes them consistent with the principles of functionalist psychology.

The idea of strong artificial intelligence, which asserts that machines might in reality have mental states if they were to acquire the appropriate level of functional organization, is supported by the theory of functionalism as well. This theory adds support to the concept of strong artificial intelligence. An artificial intelligence system that is capable of perceiving, reasoning, learning, and acting in a manner that is functionally equivalent to human cognition is able to possess beliefs, wants, and even consciousness, according to this point of view. This is because the system is able to perceive, reason, learn, and act in a manner that is like human

cognition. Behaviourism, which reduces mentality to observable behaviour alone, and identity theory, which ties mental states entirely to biological brains, are two approaches that stand in contrast to this viewpoint. Both of these perspectives seek to explain mental states by relating them to bodily brains.

Semantic networks, conceptual graphs, frames, rule-based systems, and procedural models are some examples of the representational and computational techniques that are utilised extensively by artificial intelligence (AI), in addition to logical reasoning strategies. By providing methods that are both practical and versatile for encoding and manipulating knowledge, these approaches make it possible for machines to deal with ambiguity, context, and the processing of huge amounts of information. In this regard, artificial intelligence systems frequently outperform both the neural network (NN) framework and many Western epistemological traditions. These traditions have a tendency to place an excessive amount of stress on determining the identity of the knower or the unique location where knowledge is acquired. Despite the fact that such difficulties are significant from a philosophical standpoint, it is possible that they are of minimal value to advanced artificial intelligence systems, which are able to function efficiently without a unified, conscious subject of knowledge.

Western epistemology was responsible for the development of a significant number of the conceptual tools that enabled the development of artificial intelligence. Structure, reasoning, and the evaluation of belief have been areas of emphasis throughout the past, beginning with Greek logic and continuing into current conceptions of justification. It is possible to view artificial intelligence as an extension of this endeavour, which is the endeavour to comprehend and recreate knowledge using formal means. On the other hand, ongoing discussions about meaning, consciousness, and uncertainty serve to remind us that the philosophical concerns are certainly not yet resolved.

Knock your Brain

1. Which pramāṇa is closest to AI sensor data?

A. Perception

B. Testimony

C. Postulation

D. Comparison

2. Learning from patterns in data in AI is similar to:

A. Yoga

B. Anumāna (inference)

C. Worship

D. Myth

3. Trusting information from a database in AI resembles:

A. Illusion

B. Śabda (testimony)

C. Doubt

D. Meditation

4 A major concern in both Indian epistemology and AI is:

A. Fashion

B. Validity of knowledge

C. Architecture

D. Music

5. Nyāya philosophers analyzed error in order to:

A. . Improve valid cognition

B Increase confusion

C. Reject perception

D. Avoid debate

Unleash your potential

1. Examine the theory of knowing (pramāṇa) within Indian philosophy and its significance for artificial intelligence.

2.Examine the epistemic concepts of Nyaya and Buddhist philosophy concerning artificial intelligence.

3. Analyse Behaviourism, Identity Theory, and Functionalism concerning artificial intelligence.